1 SQL Practice Exercise4: Joins, UNION, Filtering and Aggregates

1. SQL Joins:

```
INNER JOIN
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
FROM Employees AS A
INNER JOIN Projects AS B.
ON A.EmployeeID = B.EmployeeID;
```

OUTPUT:

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	102	Employee Training	50000	Ongoing
2	Alice	Smith	HR	60000	101	AI Development	100000	Completed
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Ris Management	80000	Pending

2. LEFT JOIN

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
FROM Employees AS A
LEFT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

output

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	102	Employee Trainning	50000	Ongoing
2	Alice	Smith	HR	60000	101	AI Development	100 000	Completed
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80 000	Pending

3. RIGHT JOIN

```
SELECT ProjectID,
       ProjectName,
       Budget,
       Status,
       EmployeeID
       FirstName,
       LastName,
       Department,
       Salary
FROM Employees AS A
RIGHT JOIN Projects AS B
ON A. EmployeeID = B.EmployeeID;
```

output

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
101	AI Development	100000	Completed ~~Ongoing~~	2	Alice	Smith ~~John~~	HR	60000
102	Employee Training	50000	Ongoing ~~Completed~~	1	John	Doe	IT	70000
103	Cybersecurity Audit	75000	Pending	NULL	NULL	NULL	NULL	NULL
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk Management	80000	Pending	6	Michael	Clark	Finance	80000

4. FULL OUTER JOIN

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
FROM Employee AS A
FULL OUTER JOIN PROJECTS AS B
ON A.EmployeeID = B.EmployeeID;
```

Output

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	102	Employee Training	50000	Ongoing
2	Alice	Smith	HR	60000	101	AI Development	100000	Completed
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending
NULL	NULL	NULL	NULL	NULL	103	Cybersecurity Audit	75000	Pending

3 Filtering Statements

7. Salary > 70 000

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
FROM Employees
WHERE Salary > 70000;
```

Output

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	clark	Finance	80000

8. IT or Finance

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Department IN ('IT', 'Finance');
```

Output

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

9. Project not completed

```
SELECT  ProjectID,
        ProjectName,
        Budget,
        Status
FROM    Projects
WHERE   status <> 'Completed';
```

Output:

ProjectID	ProjectName	Budget	Status
102	Employee Training	50000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

10. Budget > 70 000 and not completed

```
SELECT  ProjectID,
        Project Name,
        Budget,
        Status
FROM    Projects
WHERE   Budget > 70000 AND status != 'Completed'
```

Output

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	pending

11. Employee in New York OR Toronto sorted

```
SELECT EmployeeID, FirstName, LastName, Department, salary,
       City
FROM   Employees
WHERE  city IN ('New York', 'Toronto')
ORDER  BY salary DESC;
```

output

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	David	Brown	IT	70000	New York

12. Top 3 highest-paid employees

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM   Employees
ORDER  BY salary DESC
LIMIT  3;
```

output

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

13. Aggregate Functions

Total salary per department

```
SELECT Department,
       SUM(salary) AS TotalSalary
FROM Employees
GROUP BY Department
ORDER BY Totalsalary;
```

Output:

Department	Totalsalary
Finance	155000
IT	142000
Sales	65000
HR	60000

14. Average Salary > 65000

```
SELECT City, ~~AverageSalary~~
       Avg (Salary) AS AverageSalary
FROM Employees
GROUP BY City
HAVING AVG (salary) > 65000;
```

Output

City	AverageSalary
New York	75000
Toronto	75000
London	72000

15 Employees per department > 1

```
SELECT Department,
       COUNT(EmployeeID) AS EmployeeCount
FROM   Employees
GROUP BY Department
HAVING COUNT(EmployeeID) > 1;
```

output

Department	Employee Count
IT	2
Finance	2

16. Project per status >= 2.

```
SELECT Status,
       COUNT(ProjectID) AS project Count
FROM   projects
GROUP BY status
HAVING COUNT(ProjectID) >= 2;
```

output

Status	Project Count
Completed	2
Ongoing	2
Pending	2

17. Total project budget per employee > 150 000

```
SELECT EmployeeID,
       FirstName,
       LastName
       SUM(Budget) AS TotalProjectBudget
FROM   Employees AS A
JOIN   Projects AS B
ON     A.EmployeeID = B.EmployeeID
GROUP BY A.EmployeeID, A.FirstName, A.LasName
~~HAVING SUM (B.Budget~~
HAVING SUM (Budget) > 150 000
```